DECK 8 OF 12 · FRC JAVA CRASH COURSE

Understanding motors: read the curve

Why your robot speeds up, slows down, heats up and browns out

WPILib 2026 · Java · Command-based

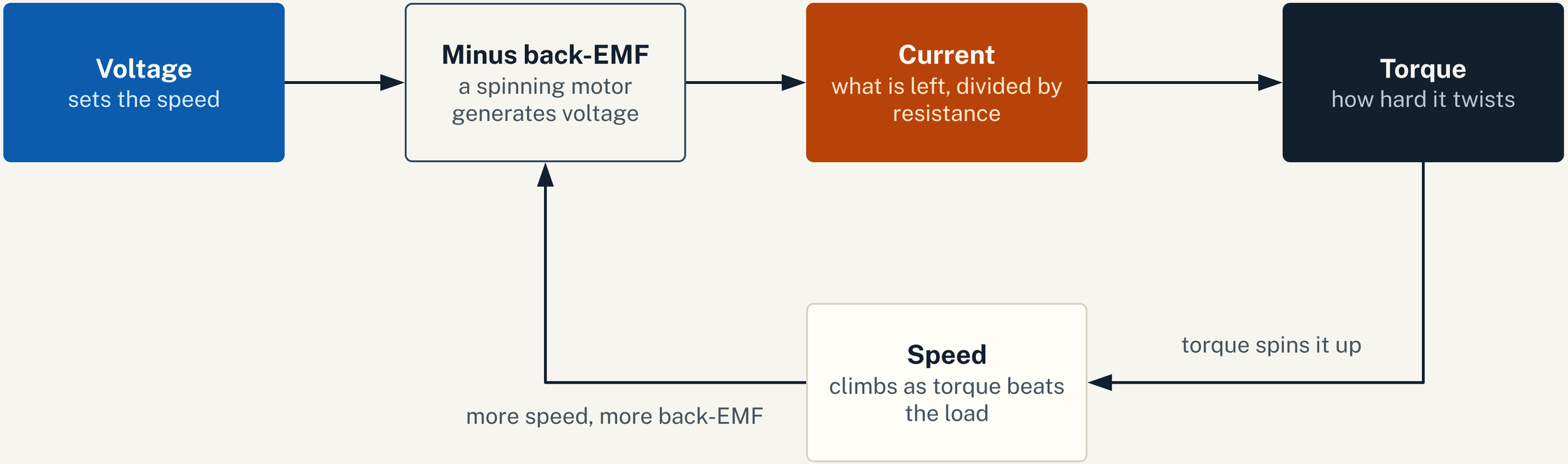
OVERVIEW

What this deck covers

- What **voltage** does to a motor, and what **current** does.
- Free speed, free current, stall torque and stall current on a motor curve.
- Working out a robot’s top speed from its gearing, by hand.
- Why a stalled motor heats up, and why the battery sags.
- **Motor current** versus **battery current**, and choosing a current limit.

MENTAL MODEL

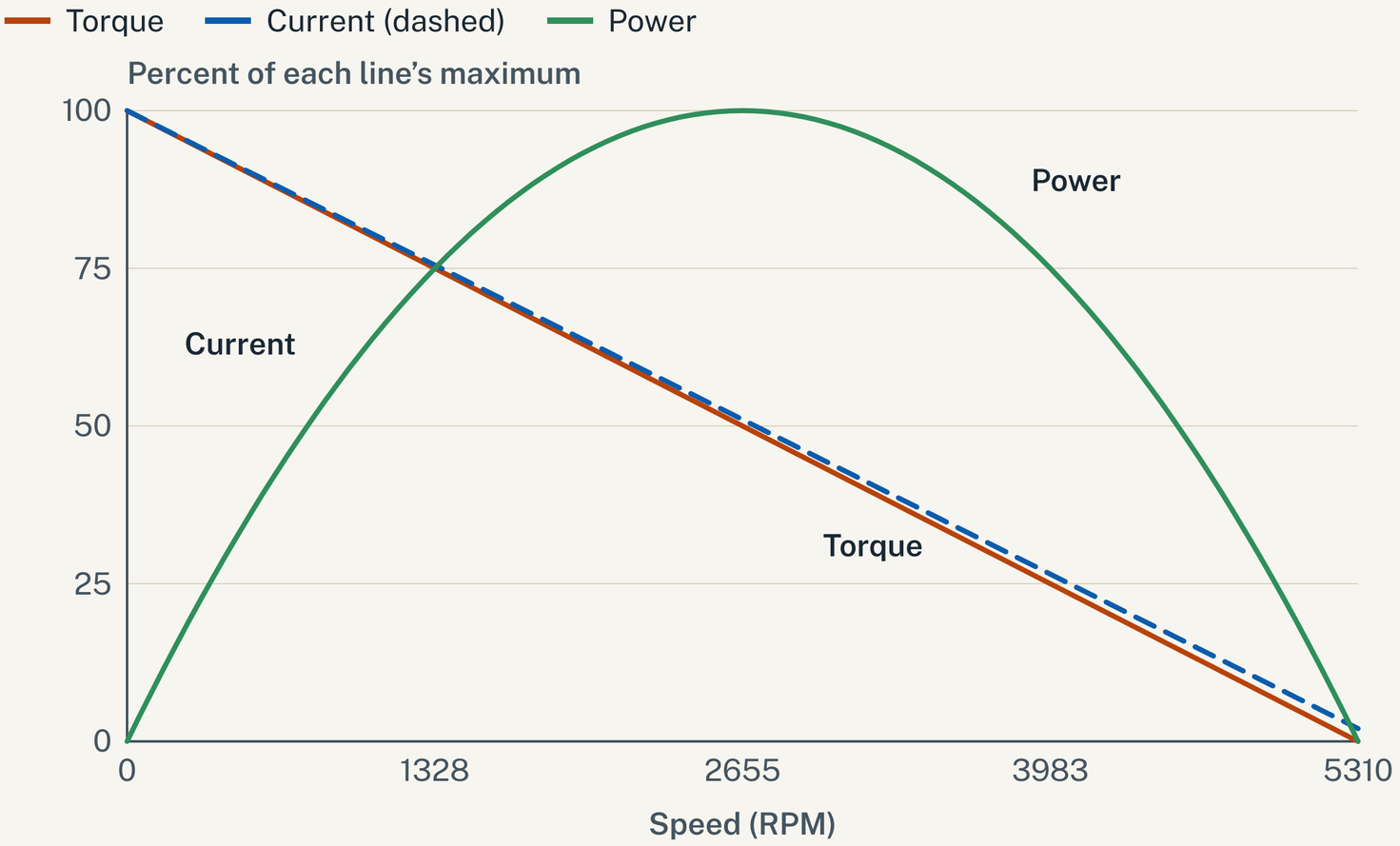
Voltage sets speed, current sets torque



Schematic. The loop settles when back-EMF leaves just enough current to carry the load.

MOTOR CURVES

Anatomy of a motor curve: a CIM at 12 V



1. **Stall: left edge**
   Stuck at 0 RPM: **2.42 Nm** of torque and **133 A**.

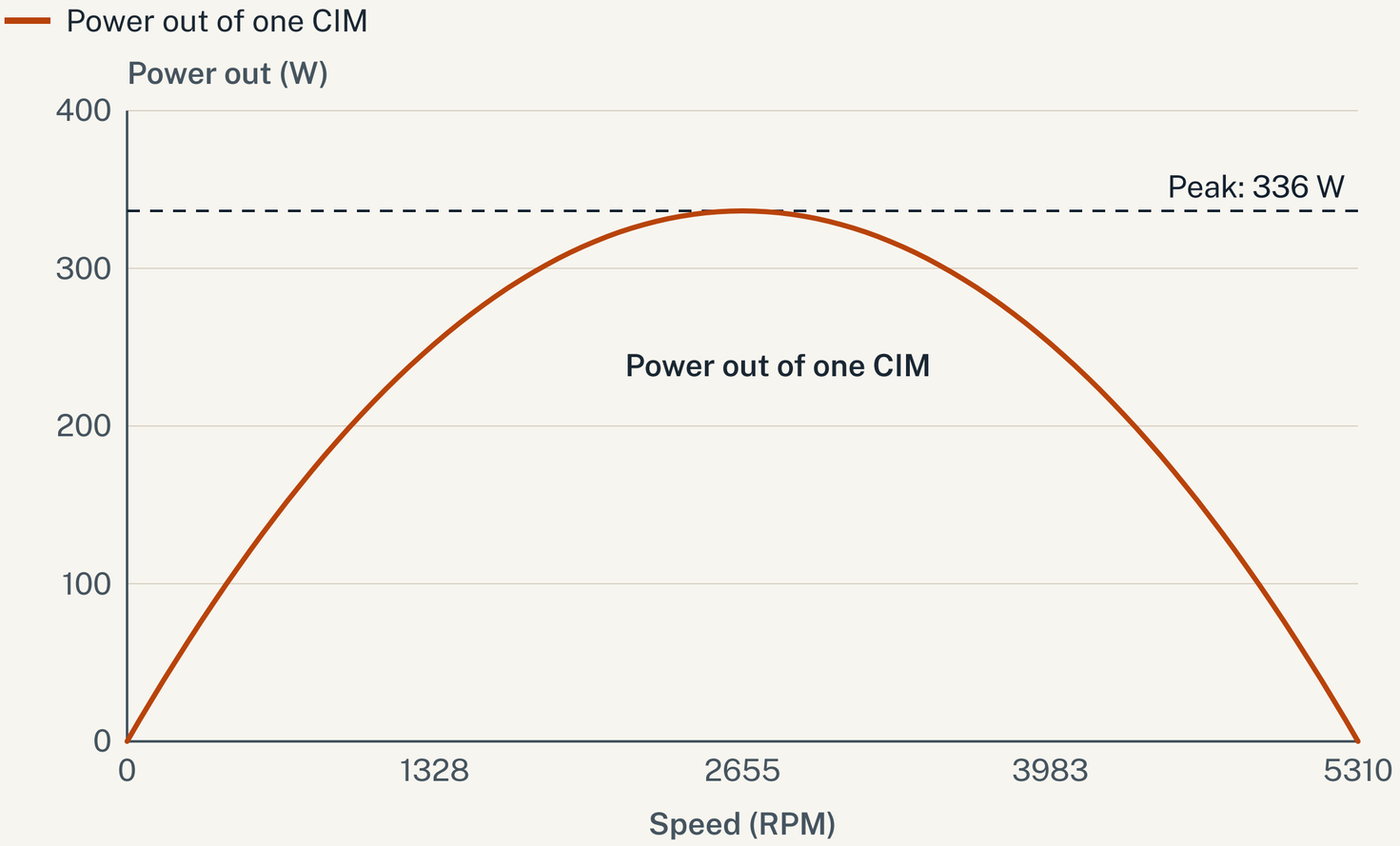
2. **Free speed: right edge**
   5310 RPM with nothing attached, and no torque left.

3. **Free current**
   2.7 A at free speed. That just beats friction.

4. **Two straight lines**
   Current makes torque, so the two fall together.

MOTOR CURVES

Peak power sits at half speed



1. **The formula**
   Power = torque × speed, with speed in radians per second (RPM × 0.105).

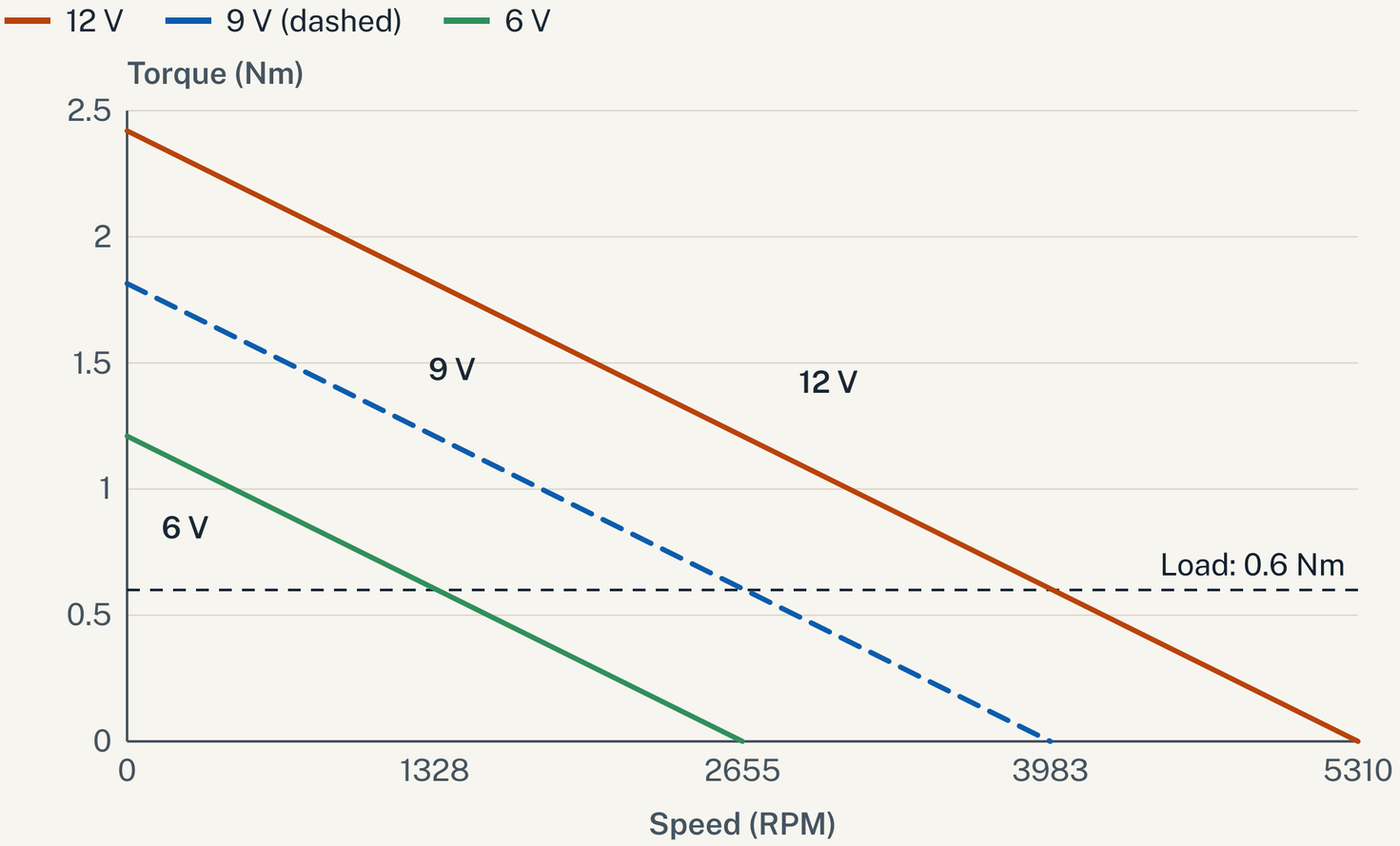
2. **Stall**
   All torque, no speed: 0 W.

3. **Free speed**
   All speed, no torque: 0 W.

4. **Half speed**
   Half of each: 1.21 Nm × 278 rad/s = **336 W**.

CONTROL THEORY

Voltage slides the curve, load picks the point



1. **Your stick**
   set(0.5) on a 12 V battery gives the motor 6 V.

2. **Same slope**
   Less voltage moves the line down. Its shape stays.

3. **Load picks the point**
   The motor runs where the lines cross: 3993 RPM at 12 V, 1338 RPM at 6 V.

4. **Same torque, same amps**
   That 0.6 Nm takes 33 A on all three lines.

MOTOR CURVES

Reading real numbers: four FRC motors

Motor	Free speed (RPM)	Free current (A)	Stall torque (Nm)	Stall current (A)	Peak power (W)
CIM	5310	2.7	2.42	133	336
NEO	5676	1.8	2.6	105	386
NEO Vortex	6784	3.6	3.6	211	639
Kraken X60	6000	2	7.09	366	1114

From WPILib DCMotor at v2026.2.1. The Kraken row is the non-FOC entry. Peak power here is stall torque × free speed ÷ 4.

GEARING

Gearing trades speed for torque: the KitBot

1. **Start at the motor**
   5310 RPM free speed and 2.42 Nm stall torque.

2. **Divide speed by the ratio**
   5310 ÷ 8.45 = **628 wheel RPM**.

3. **Multiply torque by the ratio**
   2.42 × 8.45 = **20.4 Nm** per CIM at the wheels.

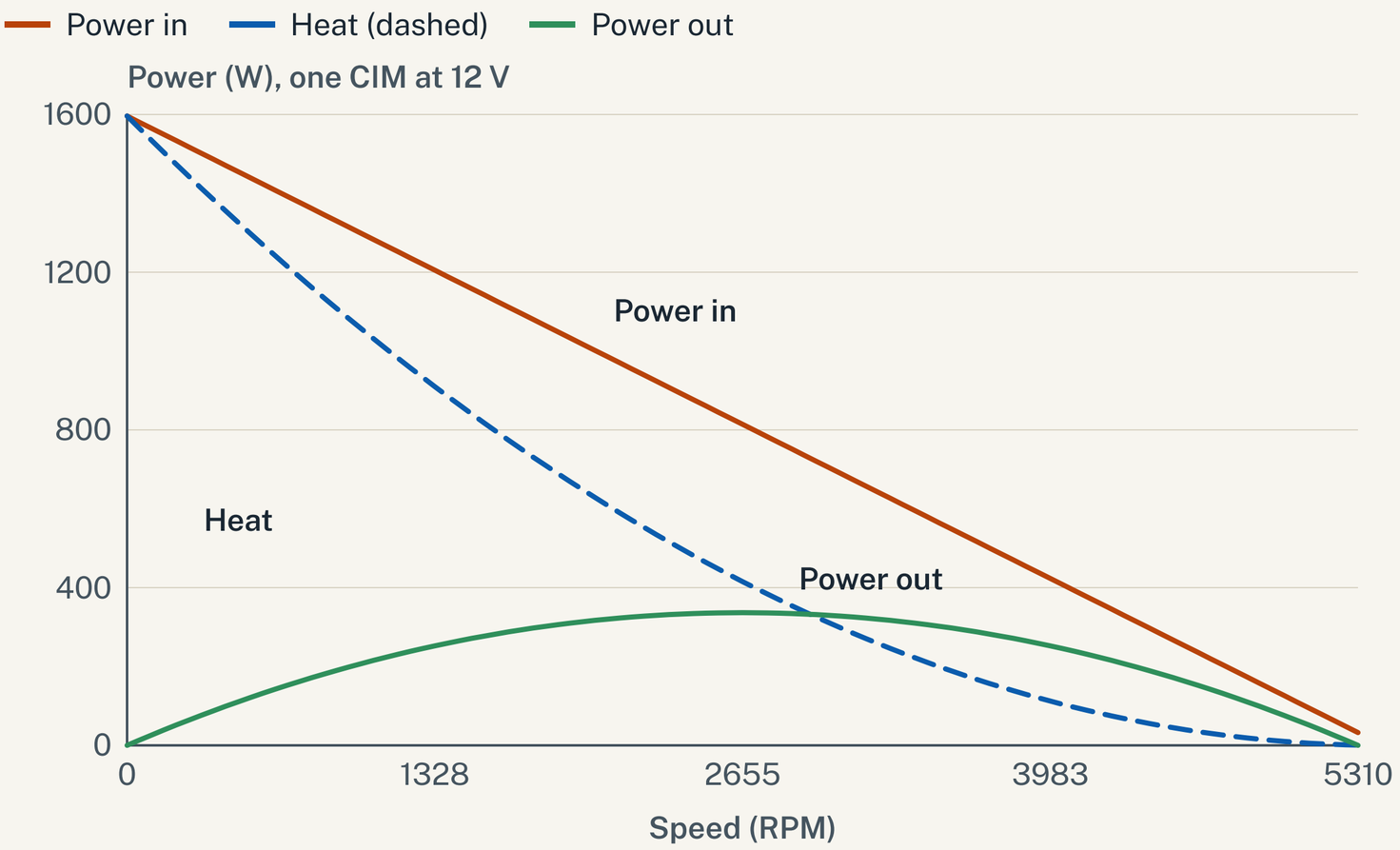
4. **One wheel turn rolls one circumference**
   6 in × π = 18.85 in, which is 0.479 m.

5. **Turns per second × distance per turn**
   628 ÷ 60 × 0.479 = **5.0 m/s** (16.5 ft/s).

6. **Compare with the real robot**
   FIRST quotes ~12.7 ft/s (3.87 m/s), 77% of this free-speed math.

HEAT

A stalled motor turns all its power into heat



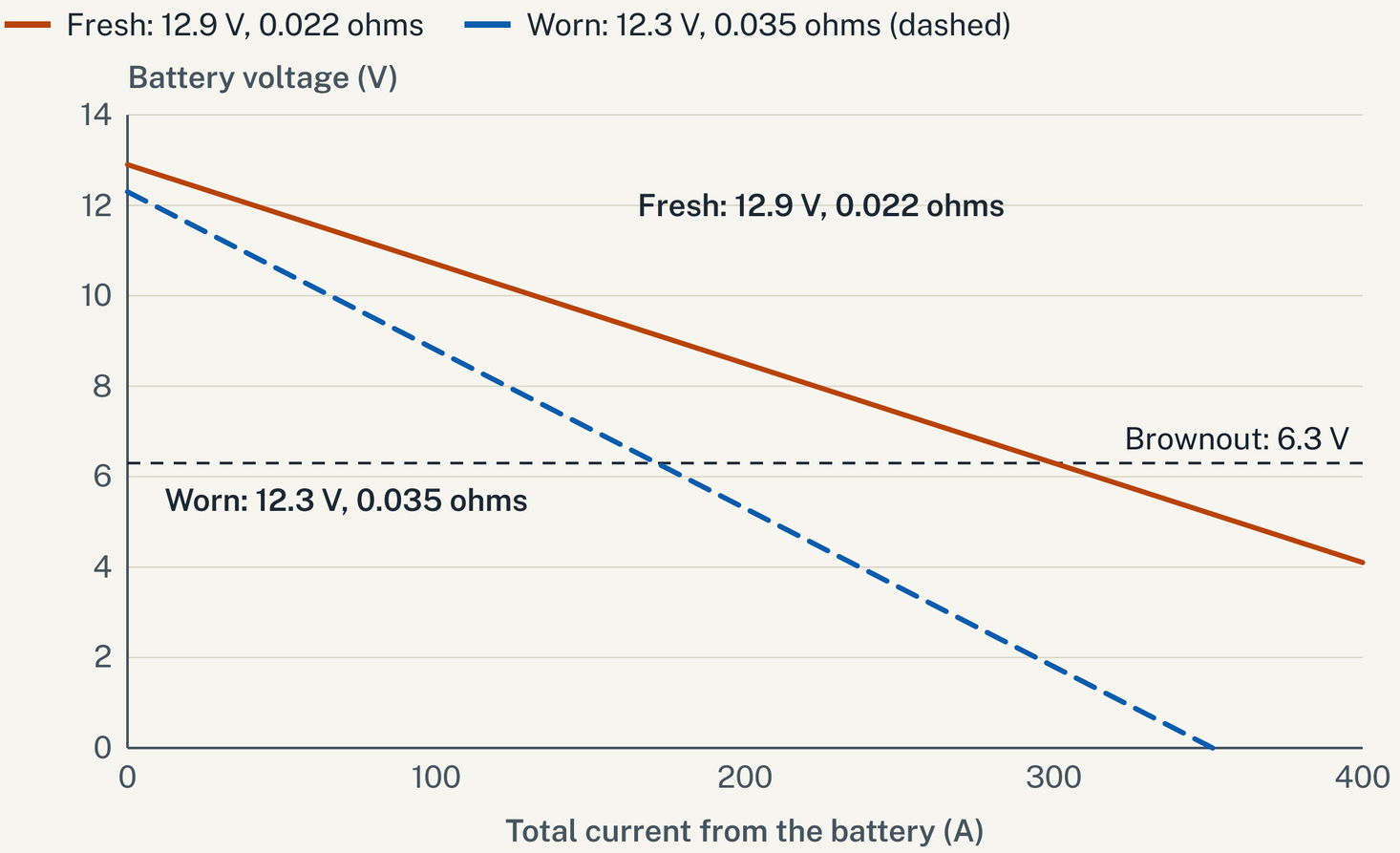
1. **At stall**
   12 V × 133 A = **1596 W** in, and nothing out.

2. **Where it goes**
   No back-EMF, so only the resistance of the windings limits the current. They heat up.

3. **At half speed**
   814 W in, 336 W out: 41% useful.

4. **Current squared**
   Half the current means a quarter of the heat.

Battery sag and brownouts



1. **Sag**
   Every amp drops voltage inside the battery.

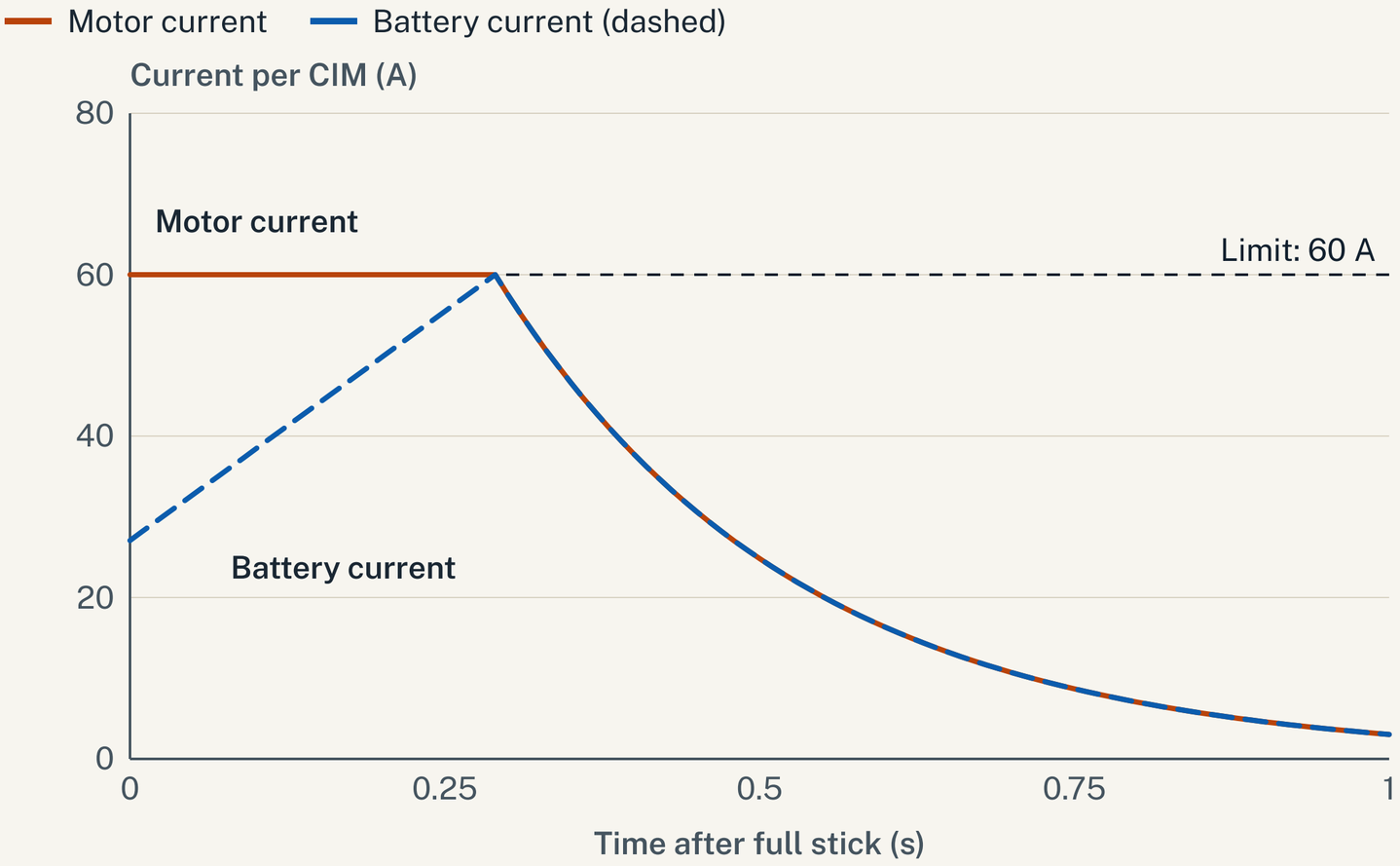
2. **Brownout**
   Below 6.3 V a roboRIO 1 cuts the motors. A roboRIO 2 defaults to 6.75 V.

3. **Full stick from a stop**
   No limit, fresh battery: 290 A, down to 6.5 V.

4. **Breakers**
   120 A main, 40 A per motor. They trip on heat after a delay, so spikes get through.

CURRENT

Motor current and battery current are different



1. **Motor current**
   In the windings. It makes torque and heat.

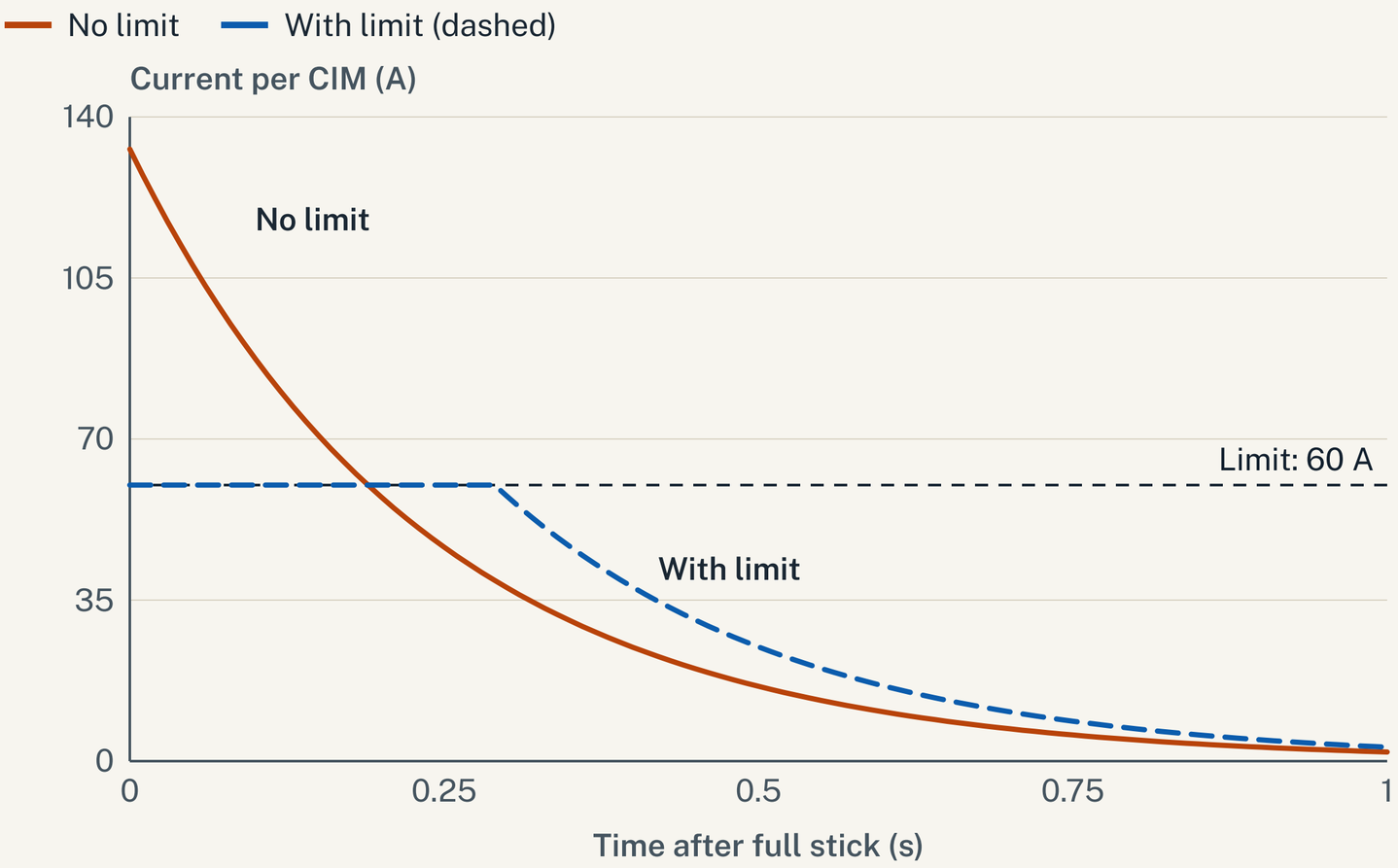
2. **Battery current**
   In the wires and breakers. It sags the battery.

3. **Same power, different split**
   12 V × 27 A in, 5.4 V × 60 A out. Same power.

4. **What the SPARK limits**
   Motor current. getOutputCurrent() reads it too.

CONTROL THEORY

What a current limit protects



1 The spike
A stopped motor has no back-EMF, so the model starts at 133 A.

2 What it protects
Breakers, battery voltage and CIM heat.

3 The cost
3 m/s takes 0.32 s, not 0.21 s.

4 Measure it
This is a model. Log getOutputCurrent() during a short, hard push.

CURRENT

Choosing a limit: stop where the wheels slip

DriveLimits.java

```
  /** Motor current where the wheels start to slip. */
  public static double slipAmps() {
1   double forceN = WHEEL_GRIP * ROBOT_MASS_KG * GRAVITY;
2   double wheelNm = forceN * WHEEL_RADIUS_METERS;
2   double motorNm = wheelNm / (DRIVE_MOTORS * GEARING);
3   return DCMotor.getCIM(1).getCurrent(motorNm);
  }

  public static SparkMaxConfig commonConfig() {
    SparkMaxConfig config = new SparkMaxConfig();
4   config.smartCurrentLimit(CURRENT_LIMIT_AMPS);
    config.idleMode(IdleMode.kBrake);
    return config;
  }
```

1. **Grip sets the force**
   Grip × weight. Our 1.0 and 50 kg are assumptions.

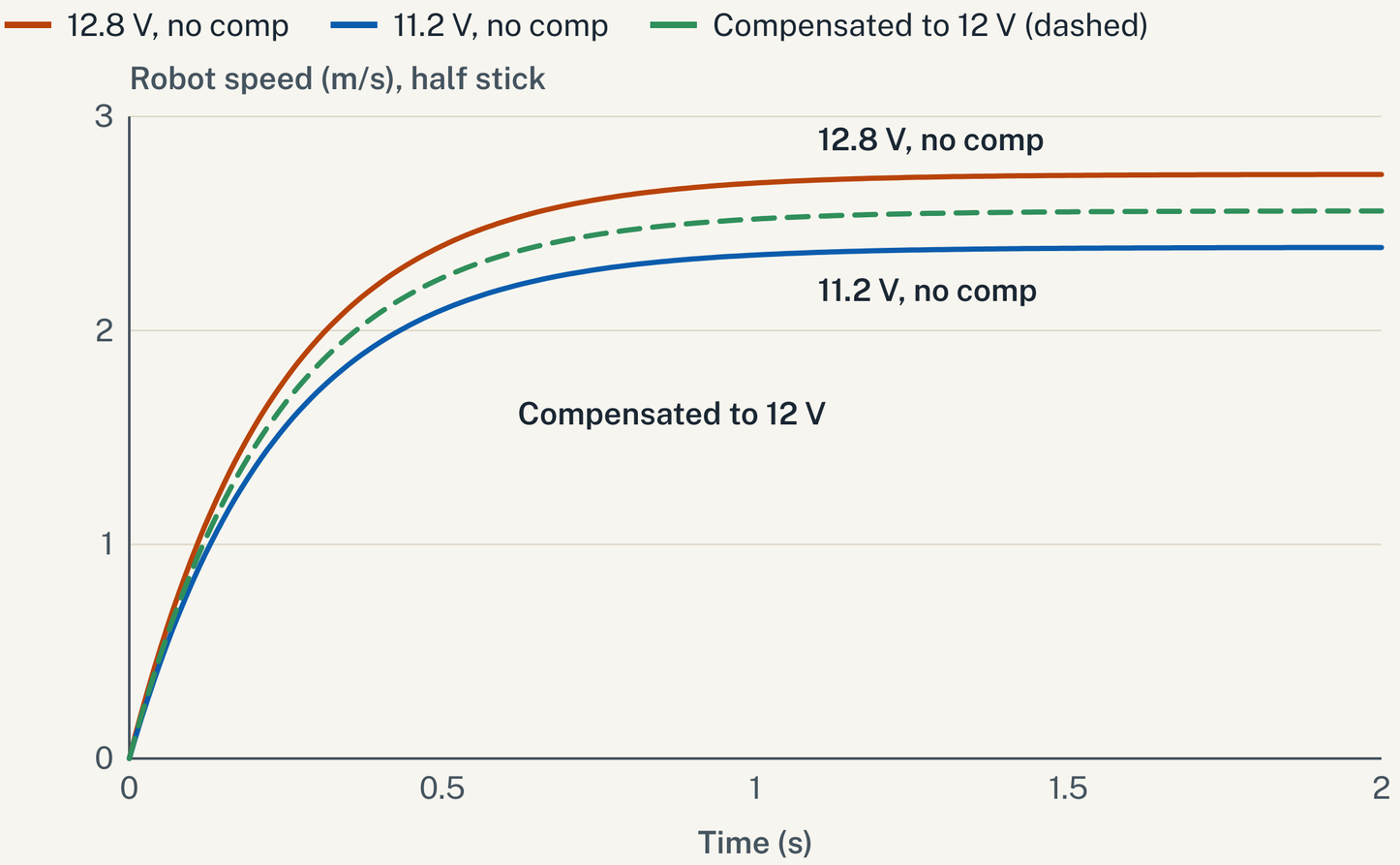
2. **Back through the gearing**
   Shared by 4 CIMs, divided by 8.45.

3. **Torque to amps**
   61 A per CIM. More just spins the wheels.

4. **Set the limit**
   FIRST’s 60 A is close to our number.

CONTROL THEORY

Voltage compensation makes output repeatable



1. **Without it**
   set(0.5) means half of whatever the battery has right now.

2. **60 cm apart**
   That is the gap after 2 s between these two batteries.

3. **With it**
   The SPARK scales output to 12 V, so both batteries match.

4. **The cost**
   A little top speed on a full battery. It can’t add volts.

COMMON MISTAKE

Comparing a motor limit with a PDH reading

What you see	The motor is stalled at a 60 A limit, but its PDH channel shows about 27 A. One of them looks wrong.
Why it happens	getOutputCurrent() is motor current. The PDH measures battery current. At low speed they are far apart.
The fix	Compare getOutputCurrent() with your limit, and the PDH with your breaker. Log both.

```
SmartDashboard.putNumber("Drive/MotorAmps", leftLeader.getOutputCurrent());
SmartDashboard.putNumber("Drive/AppliedOutput", leftLeader.getAppliedOutput());
SmartDashboard.putNumber("Drive/PdhChannelAmps", pdh.getCurrent(LEFT_LEADER_PDH_CHANNEL));
SmartDashboard.putNumber("Drive/BatteryVolts", RobotController.getBatteryVoltage());
```

BEST PRACTICE

A limit for every motor

> **Give every motor its own limit, chosen for its job.**

REV suggests 40 to 60 A for a NEO and 20 to 40 A for a NEO 550. FIRST uses 60 A on KitBot CIMs. In REV’s stall test a NEO 550 lasted about 27 s at 40 A, and about 2 s at 80 A, the SPARK default.

WPILIB

Checking your math with DCMotor

MotorCheck.java

```
public static void print() {
  // withReduction() applies the gearing to the motor's numbers.
  DCMotor side = DCMotor.getCIM(2).withReduction(GEARING);
  DCMotor cim = DCMotor.getCIM(1);

  double topSpeed = side.freeSpeedRadPerSec * WHEEL_RADIUS_METERS;
  double oneStalled = cim.getCurrent(0, NOMINAL_VOLTS);

  System.out.printf("Top speed, free-speed math: %.2f m/s%n", topSpeed);
  System.out.printf("Stall current: %.0f A per CIM, %.0f A for four%n",
      oneStalled, DRIVE_MOTORS * oneStalled);
}
```

withReduction() gears the motor down, so freeSpeedRadPerSec is now wheel speed. Radians per second × radius = metres per second.

EXERCISE

Predict the KitBot, then check it

1. Work out top speed at **8.45:1** from the curve.
2. Now do **10.71:1**, another ratio WPILib lists for the KitBot chassis.
3. Write down stall current at 12 V for one CIM, then for all four.
4. Call MotorCheck.print() in RobotContainer, then run WPILib: Simulate Robot Code.
5. Set GEARING to 10.71 and run it again. Compare with your paper.

Done when: your paper matches the console to two digits.

RECAP

What this deck covered

- **Voltage sets speed, current sets torque**, and the load decides the current.
- Free speed, free current, stall torque and stall current on any motor curve.
- Top speed by hand, and why peak power sits at half speed.
- Stall heat, battery sag and brownouts.
- Motor current versus battery current, choosing a limit, and checking it in logs.

Next: Deck 9: driver feel, feedforward and where loops run.

REFERENCE

Official docs

Brownouts and current draw

docs.wpilib.org

SPARK MAX current limits

docs.revrobotics.com

Locked rotor testing

docs.revrobotics.com

DCMotor Java docs

github.wpilib.org